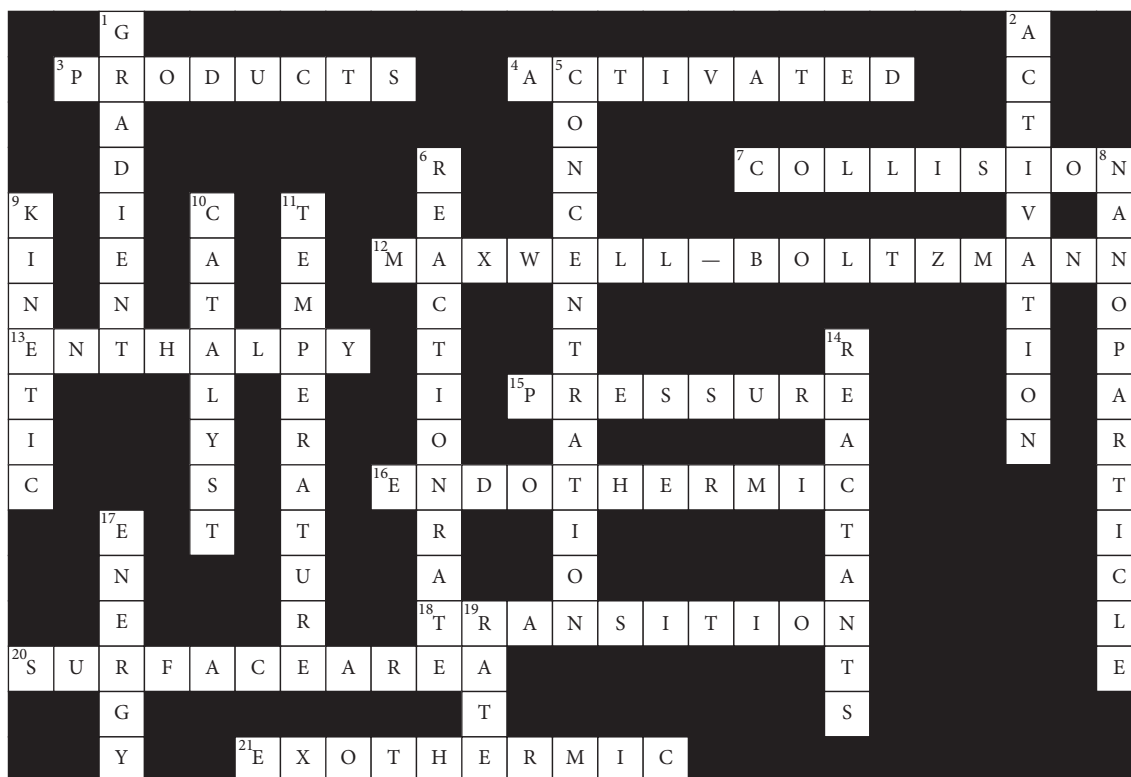


ACTIVITY SHEET ANSWERS**Chapter 9 Suggested answers****Prior knowledge**

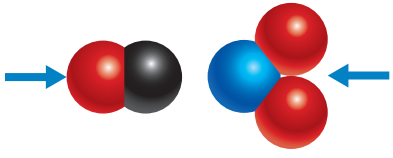
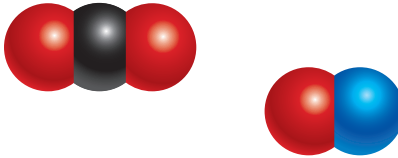
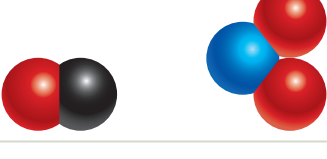
- 1 Existing chemicals are changed into new ones.
- 2 Reactants and products
- 3 They are arranged into new combinations.
- 4 Chemical bonds
- 5 Bonds are broken and new ones form.
- 6 Absorbed
- 7 Released
- 8 Energy is absorbed overall.
- 9 More energy is needed to break bonds than is released when new ones form.
- 10 Energy is released overall.
- 11 More energy is released when new bonds form than is required to break bonds.
- 12 An exothermic reaction that requires no energy to get started
- 13 No, because they all require energy to start and keep going
- 14 Yes, if it requires no energy to get started
- 15 No, because some require energy to get going
- 16 Some activation energy
- 17 None at all
- 18 The heat energy of the reactants and products, and the activation energy
- 19 Reaction rate or rate of reaction
- 20 By measuring how fast a reactant disappears or a product forms
- 21 Changes in mass, colour, volume, pH, concentration etc.
- 22 Amount of reactant consumed or product formed against time
- 23 Reaction rate at a particular point of time
- 24 Chemical particles must collide to break bonds and form new ones.
- 25 Frequency, force and orientation
- 26 Increases it because particles collide more forcefully and frequently

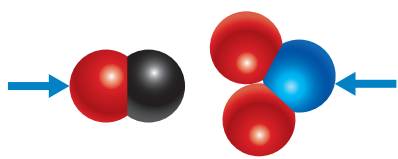
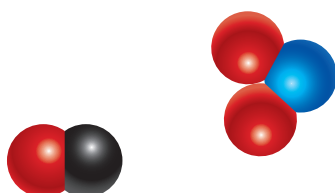
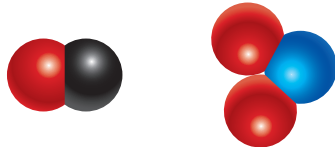
- 27 Increases it because particles collide more frequently
- 28 Increases it because more solid particles are exposed to collisions
- 29 Increases it because less activation energy is required
- 30 They are in their original state.

Activity sheet 9.1



Activity sheet 9.3

Sufficient energy at the correct orientation 	Result 
Correct orientation but insufficient energy 	Result 

Sufficient energy but incorrect orientation 	Result 
Insufficient energy and incorrect orientation 	Result 

Activity sheet 9.4

- Effect: Heating reactants would increase the rate of a reaction.

Explanation: The particles would have more kinetic energy on average and be moving faster. This would result in more frequent collisions and as more of those collisions would have sufficient energy, more products would be formed.

- Effect: Using a powdered form of a solid reactant would increase the rate of a reaction.

Explanation: More of the solid's particles would be exposed to collisions by dissolved or gaseous particles, which would result in more collisions and therefore more product.

- Effect: Increasing the pressure on a gas reactant would increase the rate of a reaction.

Explanation: Increasing the pressure on a gas reactant in effect increases its concentration per unit volume. Using a more concentrated reactant would result in more frequent collisions between particles and, as a result, more successful collisions.

- Effect: Cooling a liquid reactant would decrease the rate of a reaction.

Explanation: The particles would have less kinetic energy on average and be moving more slowly. This would result in less frequent collisions and as fewer of those collisions would have sufficient energy, less product would be formed.

- Effect: Increasing the concentration of a liquid reactant would increase the rate of a reaction.

Explanation: Increasing the concentration means there are more particles per unit volume, therefore more collisions occur, resulting in more product being formed.

- Effect: Adding more water to dissolved reactants would decrease rate of a reaction.

Explanation: Adding more water effectively reduces the concentration of the dissolved reactants, which results in fewer collisions and less product being formed.

- Effect: Adding a catalyst would increase the rate of a reaction.

Explanation: Adding a catalyst reduces the required activation energy and so a greater proportion of collisions will have sufficient energy to initiate the activation complex, therefore more product will be formed.

- 8 Effect: Increasing the volume of a gas reactant would decrease the rate of a reaction.

Explanation: Increasing the volume reduces the concentration per unit volume, resulting in fewer collisions between particles, and so fewer successful collisions forming products.

Activity sheet 9.5

- The cylinder was upside down to collect the gas accumulating above the water because the gas is less dense than water. It was filled with water initially to ensure no air was present.
- Bubbling the gas through limewater would turn the liquid cloudy if the gas was carbon dioxide.
- The reactants are CaCO_3 and HCl .
The products are CO_2 , H_2O and CaCl_2 .
- Calcium carbonate + hydrochloric acid $\rightarrow$ calcium chloride + carbon dioxide + water
 $\text{CaCO}_3(\text{s}) + \text{HCl}(\text{aq}) \rightarrow \text{CaCl}_2(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$
- Student's own graph
- The rate for the chips was reasonably rapid to start off with, then steadily dropped off until the reaction ceased. The rate for the powder was very rapid initially then dropped off earlier than for the chips, and the reaction ceased earlier too.
- The reaction using powdered calcium carbonate had a faster reaction rate.
- The reaction rate for the powder was much faster than for the chips because of the increased surface area. More calcium carbonate particles were exposed to collisions with acid particles, so more successful collisions forming products occurred.
- As the reactants were being consumed rapidly, their concentrations dropped more quickly, thus that reaction slowed first.
- There may have been an error in weighing exactly 2 g of marble chips, which are relatively large.

Chapter 9 Revision sheet

- a $\rightarrow$ G, b $\rightarrow$ M, c $\rightarrow$ T, d $\rightarrow$ E, e $\rightarrow$ B, f $\rightarrow$ A, g $\rightarrow$ P, h $\rightarrow$ W, i $\rightarrow$ C, j $\rightarrow$ N, k $\rightarrow$ Q, l $\rightarrow$ D, m $\rightarrow$ O, n $\rightarrow$ L, o $\rightarrow$ F, p $\rightarrow$ S, q $\rightarrow$ J, r $\rightarrow$ I, s $\rightarrow$ R, t $\rightarrow$ H, u $\rightarrow$ V, v $\rightarrow$ K, w $\rightarrow$ U
- Activation energy* refers to the minimum amount of energy required to break the bonds of the reactants. An *activation complex* is the intermediate product in a reaction when old bonds have broken but new ones have yet to form.
 - An *exothermic reaction* releases heat energy to the environment because the products have less enthalpy than the reactants. An *endothermic reaction* absorbs heat energy from the environment because the products have more enthalpy than the reactants.
 - Enthalpy* refers to the total energy possessed by a chemical substance. *Kinetic energy* refers to the energy possessed by moving particles.
 - An *energy profile diagram* shows the relative energies of the reactants, the activated complex and the products during the progress of a reaction. A *reaction progress graph* shows the rate at which a reaction is occurring by plotting how fast a reactant is being used up or a product is being formed.

- 3 a The steepness of the graph curve indicates how fast a reaction is proceeding. A steep slope (or gradient) indicates that the reaction is fast, a flatter slope indicates the reaction is slow.
- b The reaction rate as shown by the gradient of the curve was greatest at the start of the reaction then gradually slowed to zero after about 15 s has elapsed.
- c The high concentration of the hydrochloric acid at the start of the reaction causes the very fast initial reaction rate, but as the reaction proceeded the acid concentration fell and there was less magnesium, so there were fewer collisions between acid particles and magnesium atoms.
- 4 a Reactant particles must collide with sufficient force so that their combined energy exceeds the activation energy required for the reaction.
- b The colliding particles must meet at the right orientation to allow existing bonds to be broken and new bonds to be formed.
- c The more frequently collisions occur, the more successful collision there will be and the faster the reaction.

Action	Effect on reaction rate	Collision theory explanation
Heating the reactants	<i>Increases</i>	<i>Heating increases the energy and frequency of collisions as the reactants particles are moving faster</i>
Cooling the reactants	<i>Decreases</i>	<i>Slower moving particles collide less frequently and with less energy</i>
Using a more concentrated reactant	<i>Increases</i>	<i>As there are more reactant particles, the frequency of collisions will increase</i>
Adding more water to a reaction	<i>Decreases</i>	<i>There will be fewer reactant particles per unit volume, so fewer collisions will occur</i>
Increasing the pressure in a reaction vessel	<i>Increases</i>	<i>Forcing reactant particles closer together means that collisions will be more frequent</i>
Using a powdered solid rather than a lump	<i>Increases</i>	<i>The powder has many more particles exposed to collisions, so collisions occur more frequently</i>
Dissolving a solid reactant first	<i>Increases</i>	<i>The reactants will collide much more frequently when both are in a dissolved state</i>
Adding a catalyst	<i>Increases</i>	<i>As the activation energy is reduced, more collisions are successful</i>

- 6 a The right reaction is endothermic because the products have more enthalpy than the reactants so energy must have been absorbed from the environment.
- b The left reaction is exothermic because the reactants have more enthalpy than the products so energy must have been released into the environment.
- c The right (endothermic) reaction, as more than twice the energy needed to start the left reaction is required to start the right reaction.
- d The energy required to break the bonds of the reactants is more than the energy released when the product bonds form.
- e The energy required to break the bonds of the reactants is less than the energy released when the product bonds form.

- 7 a At the higher temperature, a greater proportion of the reactant molecules will have higher speeds and therefore more kinetic energy. The particles will therefore collide more frequently, but more importantly more particles will have a greater amount of energy than the activation energy.
- b By providing an alternative reaction pathway, the catalyst lowers the activation energy and therefore more reactant particles will have more energy than the activation energy.
- c No, it does not alter the distribution of speeds/kinetic energies at all.